

Task 1 - Natural Language Processing Skill (19.6.26)

```
import nltk
```

```
from nltk.tokenize import word_tokenize
```

```
from nltk.corpus import stopwords
```

```
from nltk.stem import PorterStemmer, WordNetLemmatizer
```

```
from nltk.probability import FreqDist
```

```
from nltk import pos_tag
```

```
resume_text = """
```

```
John Smith is a Python Developer with experience in Machine Learning,  
Data Analysis, SQL, and Web Development. He has worked on NLP projects  
and developed AI-based applications.
```

```
"""
```

```
tokens = word_tokenize(resume_text)
```

```
tokens = [word.lower() for word in tokens if word.isalpha()]
```

```
print("Tokens:")
```

```
print(tokens)
```

```
stop_words = set(stopwords.words('english'))
```

```
filtered_words = [word for word in tokens if word not in stop_words]
```

```
print("\nAfter Stopword Removal:")
```

```
print(filtered_words)
```

```
stemmer = PorterStemmer()
```

```
stemmed_words = [stemmer.stem(word) for word in filtered_words]
```

```
print("\nStemmed Words:")
```

```
print(stemmed_words)
```